CALCIUM METABOLISM, CALCIFICATION

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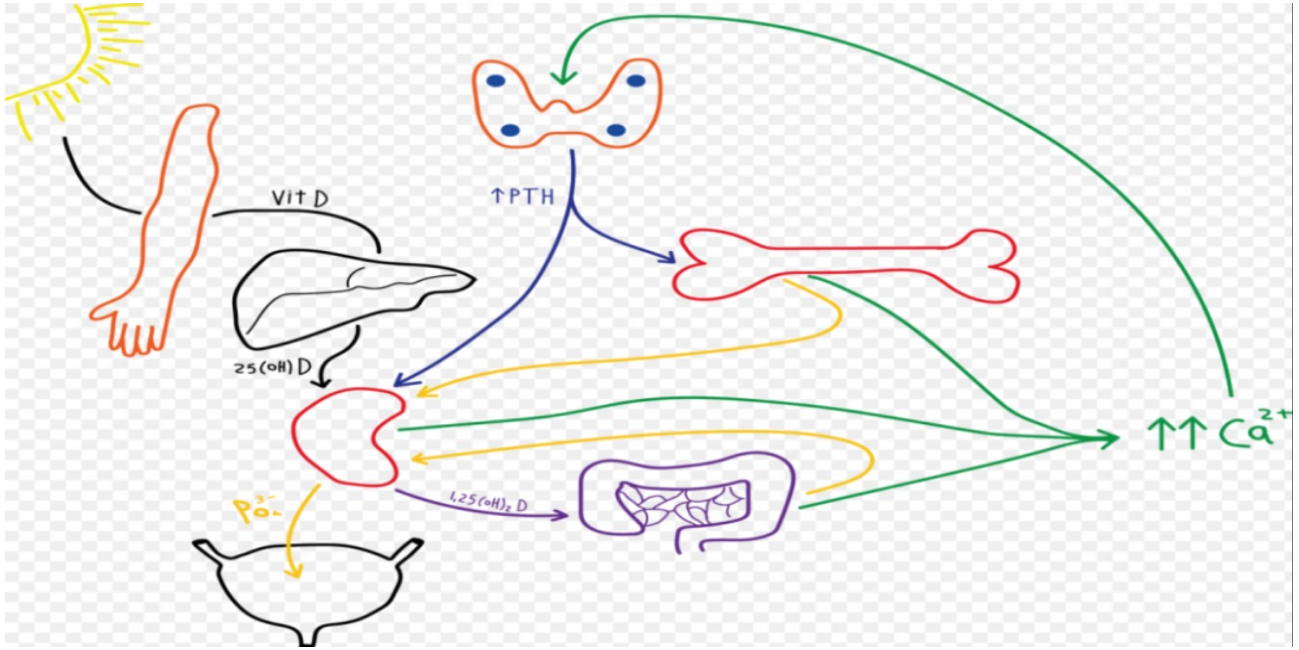
Calcium metabolism

- Calcium metabolism is the movement and regulation of calcium ions (Ca2+) in (via the gut) and out (via the gut and kidneys) of the body, and between body compartments: the blood plasma, the extracellular and intracellular fluids, and bone.
- **Bone** acts as a **calcium storage center** for deposits and withdrawals as needed by the blood via continual **bone remodelling**



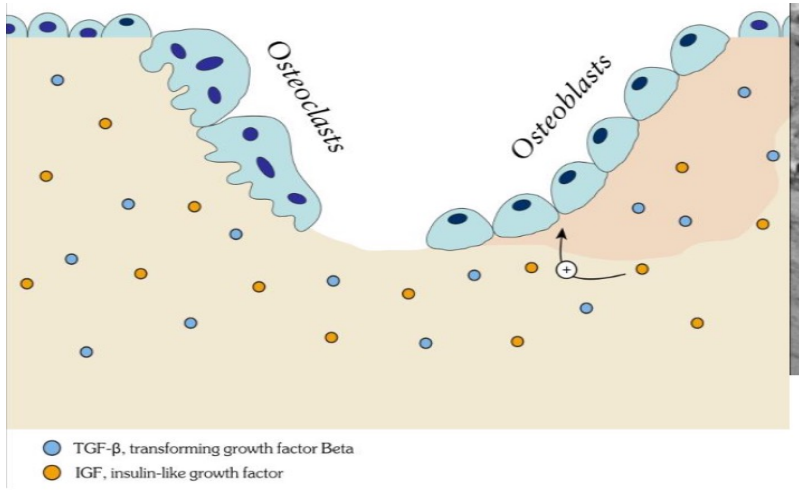
Metabolism

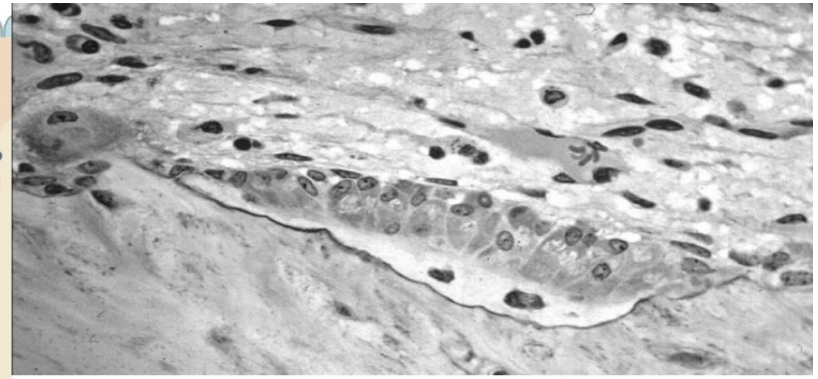
- is the set of life-sustaining chemical reactions in organisms. The three main functions of metabolism are:

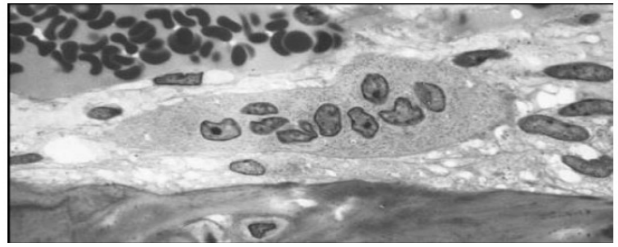
1. the conversion of the energy in food to energy available to run cellular processes;
2. the conversion of food to building blocks for proteins, lipids, nucleic acids, and some carbohydrates; and
3. the elimination of metabolic wastes

Calcium Metabolism

- In the first year of life, almost 100% of the skeleton is replaced.
- In adults, remodelling proceeds at about 10% per year.
- An imbalance in the regulation of bone remodelling's two sub-processes, bone resorption and bone formation, results in many metabolic bone diseases, such as osteoporosis.



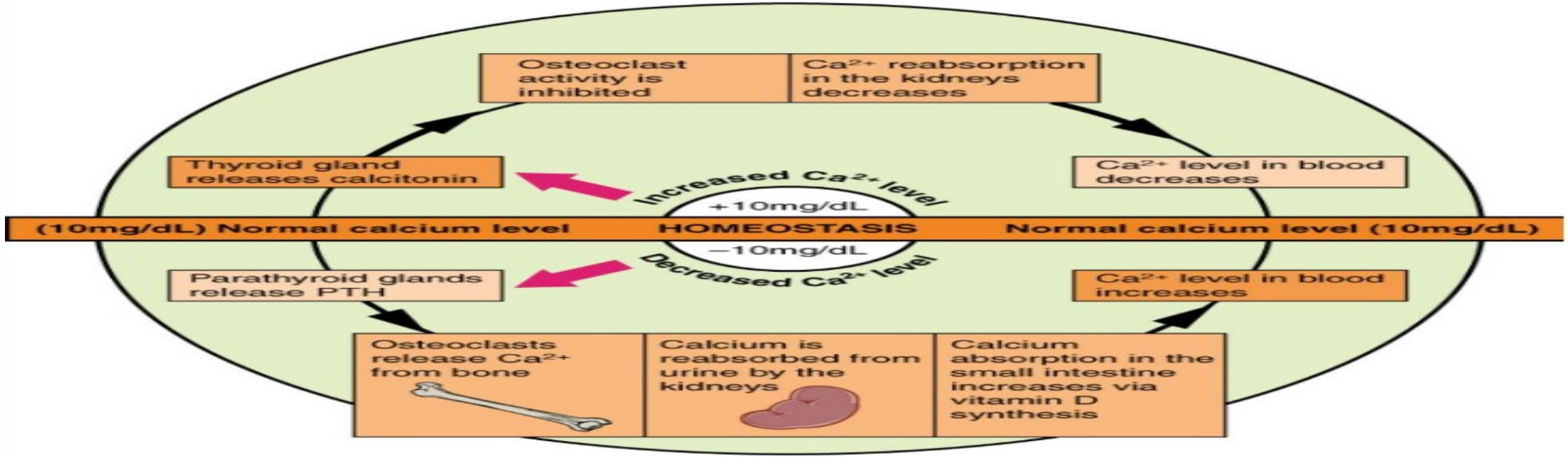




Calcium Metabolism

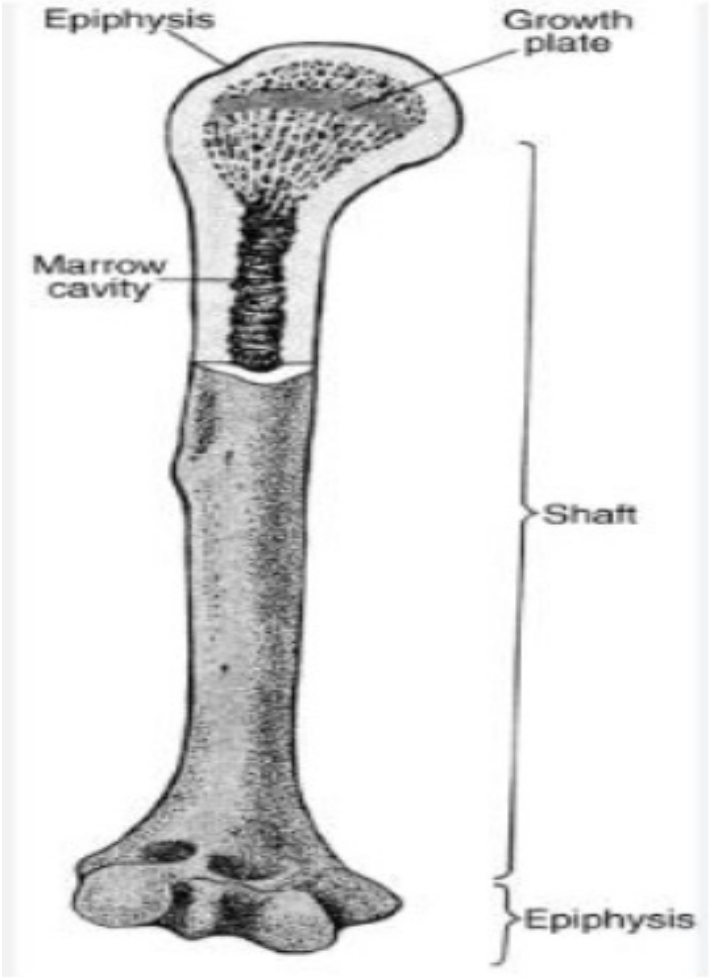
- The body regulates calcium homeostasis with two pathways;

1. one is signaled to turn on when blood calcium levels drop below normal and
2. one is the pathway that is signaled to turn on when blood calcium levels are elevated



Calcium Metabolism

- **Blood concentration:**
- The concentration of calcium ions inside cells (in the intracellular fluid) is more than 7,000 times lower than in the blood plasma (i.e. at <0.0002 mmol/L, compared with 1.4 mmol/L in the plasma)
- Calcium acts structurally as supporting material in bones as calcium hydroxyapatite ($Ca_{10}(PO_4)_6(OH)_2$).



Calcium Metabolism

- **Intestinal absorption**
- The normal adult diet contains about 25 mmol of calcium per day.
- Only about 5 mmol of this is absorbed into the body per day.
- **Excretion:**
- **Intestine**
- Most excretion of excess calcium is via the bile and feces, because the plasma calcitriol levels (which ultimately depend on the plasma calcium levels) regulate how much of the biliary calcium is reabsorbed from the intestinal contents.
- **Kidneys**
- Urinary excretion of calcium is normally about 5 mmol (200 mg) /day. This is less in comparison to what is excreted via the feces (15 mmol/day).

Calcium Metabolism

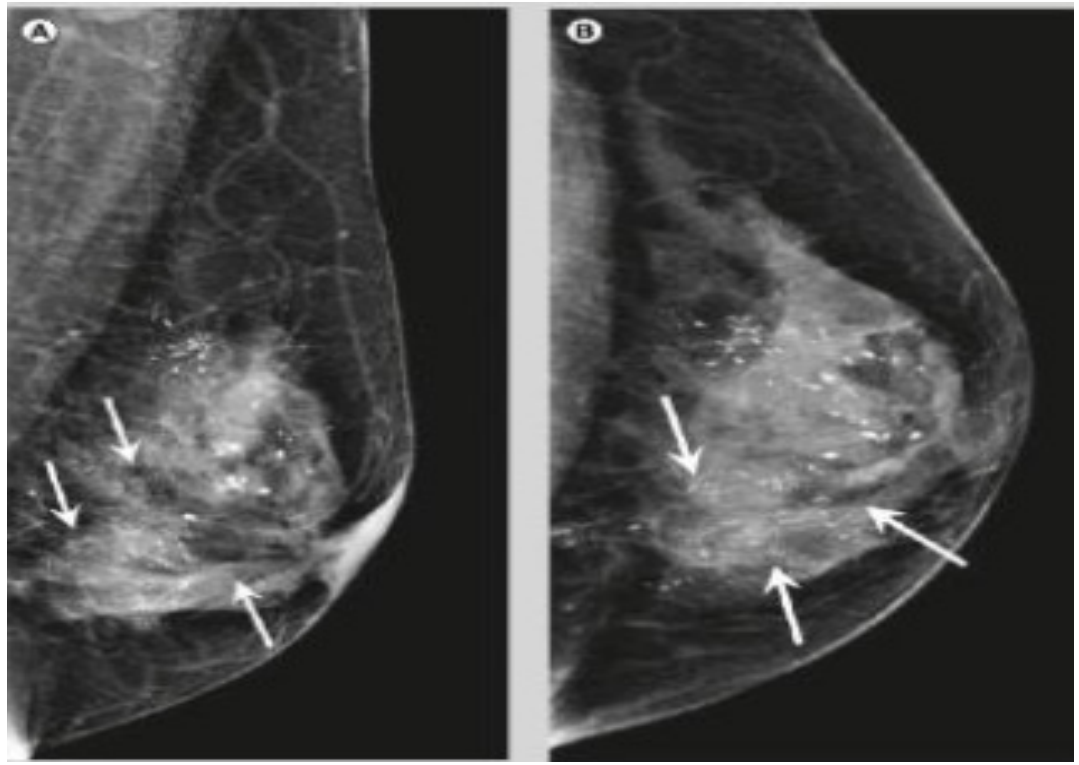
- Calcification occurs when a build-up of excess calcium occur in the body. There are many different types of calcium deposits. They can form all over the body, including in the soft tissues, arteries and organs. Some calcium deposits are normal, but others can signal a disease process.

Types of Calcification

- **Skin calcification:** is called calcinosis cutis. They can occur anywhere in the body, most commonly on the fingertips. They can also occur on the face (including the eyelids) and joints, such as elbow and knee.

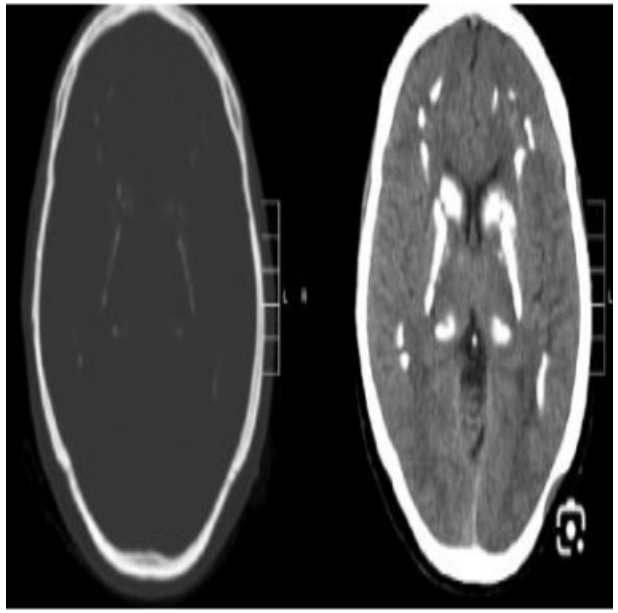
Types of Calcification

- **Breast calcification:** Calcium deposits in breasts usually don't cause any symptoms. They're too small to feel, and are usually seen on a mammogram as small, bright white spots.
- **Shoulder calcification:** Calcium deposits in shoulders can be either asymptomatio or they cause symptoms, such as pain and discomfort on movement with/without a decreased range of motion.



Types of Calcification

- **Brain calcification** usually occur in the blood vessels commonly in the basal ganglia which control movement in the body.
- **Kidney calcification** is called nephrocalcinosis. This usually occurs in the inner part of your kidney (the renal medulla). Most of the time, the condition affects both kidneys.
- **Arteries**: Calcium deposits can cause them to stiffen. This increases the risk of cardiovascular disease. Coronary artery calcification increases with age, with deposits found in 90% of men and 67% of women older than 70.



Symptoms and Causes

- **Inflammation:** Inflammation can cause tissue damage. This can cause the body to release proteins that bind calcium in clumps.
- **Calcium metabolism disorders**: These types of disorders can lead to hypercalcemia (raised Calcium level in the blood).
- **Autoimmune disorders:** that affect the skeletal system or connective tissues can cause calcification.

Questions

1. What is the primary function of calcium metabolism?
   a) Conversion of food to energy
   b) Conversion of food to building blocks
   c) Elimination of metabolic wastes
   d) Regulation of blood calcium levels
   e) Storage of calcium in bones
2. What is the main storage center for calcium in the body?
   a) Blood plasma
   b) Extracellular fluid
   c) Intracellular fluid
   d) Kidneys
   e) Bones
3. What percentage of the skeleton is replaced in the first year of life?
   a) 10%
   b) 25%
   c) 50%
   d) 75%
   e) 100%

Questions

4. How much calcium from the normal adult diet is absorbed into the body per day?
   a) 5 mmol
   b) 10 mmol
   c) 15 mmol
   d) 20 mmol
   e) 25 mmol
5. Where do brain calcifications commonly occur?
   a) Cerebral cortex
   b) Hippocampus
   c) Basal ganglia
   d) Cerebellum
   e) Frontal lobe
6. How does inflammation contribute to the formation of calcium deposits?
   a) Triggers release of calcium from bones
   b) Causes hypercalcemia
   c) Binds calcium in clumps
   d) Increases urinary excretion of calcium
   e) Impairs intestinal absorption of calcium